

Abhin Binu Kumar

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Quantitative Research · Financial Modelling · Sector Analysis

PROFESSIONAL SUMMARY

Mathematics and Computing graduate of the Indian Institute of Technology (IIT) Goa (CGPA 8.24/10, academic rank 5) with experience in quantitative financial analysis, multi-year financial modelling, and large-scale data analysis. Prepares feasibility and investment analysis across hospitality, agriculture, healthcare, automotive, and renewable energy, using Python and Excel for scenario analysis, break-even analysis, and debt service coverage ratio (DSCR) sensitivity. Additional experience includes C++ optimisation for operational decision systems and statistical analysis of telecom datasets of more than 6 million customer records. Academic work covers derivatives pricing, Monte Carlo simulation, time-series models, and privacy-preserving synthetic data. Currently advancing expertise in mathematical finance, stochastic modelling, and derivatives pricing through graduate study.

CORE SKILLS

Financial analysis: Financial modelling (modeling), cash-flow analysis, scenario analysis, break-even analysis, DSCR, feasibility analysis, profitability analysis, capital expenditure, investment decision support

Markets and pricing: CAPM, systematic risk, Black–Scholes, option Greeks, geometric Brownian motion, Monte Carlo simulation, hedging sensitivity, volatility, risk management, backtesting

Data and statistics: Statistical modelling, time-series analysis, exploratory data analysis, feature engineering, anomaly detection, churn analysis, forecasting, datasets of 6 million+ records

Optimisation: Branch and Bound, mixed-integer linear programming (MILP), constrained optimisation (optimization), numerical methods, finite element methods, matrix methods

Computing: Python, C++, Microsoft Excel, FreeFem

Machine learning: LSTM sequence models, generative models, differential privacy, PATE, synthetic data

Sectors: Hospitality, agriculture, healthcare, automotive, renewable energy, telecom, logistics and fleet operations

PROFESSIONAL EXPERIENCE

Quantitative Research Associate | Sep 2025 – Present

Neel Corporation, India

- Conduct quantitative financial analysis for live projects in hospitality, agriculture, healthcare, automotive, and renewable energy, supporting feasibility studies and investment decisions.
- Build and analyse multi-year financial models covering revenue, operating costs, capital expenditure, profitability, and cash-flow dynamics.
- Use Python and Excel for scenario analysis of financing assumptions, including break-even points and debt service coverage ratio (DSCR) sensitivity, and compare project viability across those cases.

Research Engineer – Optimisation and Decision Modelling | Sep 2024 – Sep 2025

Addverb Technologies, India

- Worked on large-scale optimisation and constrained decision-making for fleet management systems.
- Designed and implemented an optimal assignment algorithm using Branch and Bound, with a cost function covering distance, congestion, and fairness.
- Implemented the optimisation pipeline in C++, developing pruning strategies that replaced an MILP-based solver and improved system throughput by 18%.
- Analysed optimality–efficiency trade-offs between the Branch and Bound solver and the previous MILP approach.

Machine Learning Research Intern | May 2023 – Dec 2023

India Urban Data Exchange (IUDX)

- Designed and implemented a PATE-GAN framework by integrating Private Aggregation of Teacher Ensembles (PATE) with generative models to produce synthetic datasets with formal differential privacy guarantees.
- Evaluated privacy–utility trade-offs using statistical and distributional metrics on sensitive datasets, including finance-related data.
- Focused on privacy-aware machine learning and collaborated with researchers on responsible AI deployment.
- Presented the work at the Symposium on Data for Public Good, Indian Institute of Science (IISc) Bangalore.

Linear Algebra Tutor | Jan 2023 – Feb 2023

Unacademy

- Taught vector spaces, eigenvalues, eigenvectors, and matrix factorisation.

Data Analyst Intern | Jul 2022 – Aug 2022

6D Technologies, India

- Analysed churn behaviour in datasets of more than 6 million customers using statistical modelling, exploratory data analysis, and time-series techniques.
- Performed feature analysis and anomaly detection on large-scale telecom data using Python pipelines built for high-volume, noisy datasets.

SELECTED PROJECTS

Finite Element Methods (FEM) for Solving Partial Differential Equations | Dec 2023 – May 2024

Academic Project, IIT Goa

- Implemented Finite Element Method formulations in FreeFem to solve partial differential equations, modelling systems such as heat transfer and electrostatics.
- Applied matrix assembly, decomposition, and stability analysis to large linear systems.
- Conducted stability, convergence, and error analysis for the numerical schemes.

Algorithmic Trading System with Machine Learning | Dec 2023 – Apr 2024

IIT Goa

- Designed an end-to-end algorithmic trading system covering data ingestion, feature engineering, signal generation, execution, and risk management.
- Implemented LSTM-based sequence models for financial time series and integrated them with volatility-adjusted position sizing, stop-loss rules, and drawdown constraints.
- Built a modular Python backtesting pipeline to evaluate strategies across different market conditions.

Mathematical Finance and Asset Pricing Models | Feb 2023 – May 2023

Course project, Computational Finance: An Introduction to Option Pricing, IIT Goa

- Studied the Capital Asset Pricing Model (CAPM) and risk–return trade-offs, including the role of systematic risk in expected asset returns.
- Implemented Black–Scholes option pricing and derived analytical prices and Greeks under geometric Brownian motion.
- Developed Monte Carlo simulations for derivative pricing, validated them against closed-form solutions, and analysed option sensitivity to volatility, interest rates, and time to maturity for hedging.

EDUCATION

Bachelor of Technology in Mathematics and Computing | 2020 – 2024

Indian Institute of Technology (IIT) Goa, India

CGPA: 8.24 / 10 · Academic rank: 5th

Relevant coursework. Mathematics and finance: Linear Algebra, Probability and Statistics, Calculus, Differential Equations, Numerical Analysis, Computational Finance (Brownian motion, Itô's lemma, stochastic differential equations, Feynman–Kac, Black–Scholes), Complex Analysis, Discrete Mathematics. Computing and data: Algorithm Design, Data Structures and Algorithms, Econometric Data Science, Machine Learning, Theory of Computation, Introduction to Computing.

COURSES AND CERTIFICATIONS

Udemy and Coursera

- Introduction to Risk Management
- Introduction to Time Series and Forecasting
- Algorithmic Trading and Quantitative Analysis Using Python
- Mathematics for Machine Learning and Data Science Specialization

CONFERENCE PRESENTATION

Symposium on Data for Public Good, Indian Institute of Science (IISc) Bangalore | Nov 2023

Topic: Privacy-Preserving Synthetic Data using PATE-GAN

ACHIEVEMENTS

- Prime Minister's Scholarship Scheme (PMSS), Government of India, awarded for academic merit during undergraduate studies.
- Qualified JEE Advanced, the national entrance examination for the Indian Institutes of Technology.
- Captain, IIT Goa Football Team. Represented the institute at the Inter-IIT Sports Meet.
- State-level chess player. Secured 27th rank in the State Chess Championship.